Greenplum
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Welcome to Greenplum Performance Monitor 2.0.0.1

Greenplum Performance Monitor is an optional performance monitoring feature that administrators can install and enable for Greenplum Database 3.2.1. Greenplum Performance Monitor lets administrators view the status and performance of actively running queries and allows for detailed analysis of historical queries and system metrics. A browser-based graphical interface provides easy access to a variety of configurable views and reports on performance data.

Greenplum Performance Monitor 2.0.0.1 is the first patch release for the 2.0.0.x version. For the contents of the previous 2.0.0.0 release, see the following section:

- Greenplum Performance Monitor 2.0.0.0

Users wishing to upgrade to 2.0.0.1, see Upgrading to Greenplum Performance Monitor 2.0.0.1 for detailed upgrade instructions.

This patch release includes the contents of 2.0.0.1, and corrects the following additional issues:

Table A.1 Resolved Issues in Greenplum Performance Monitor 2.0.0.1

Issue Number	Description
4887	**Temporary query files may accumulate when Greenplum Performance Monitor is stopped** When Greenplum Performance Monitor is stopped on an otherwise active Greenplum system, query files in `$MASTER_DATA_DIRECTORY/gpperfmon/data` may not be cleaned up after the queries finish. If the monitor is stopped for a long time, the number of files in this directory may keep growing, using more and more disk space. This issue is resolved in release 2.0.0.1.
3131	**Value for cpu_idle is greater than 100%** Because the underlying function for retrieving CPU usage provides percentage values per physical CPU, the total for a dual-CPU system is 200%. This can result in CPU percentage metrics greater than 100%. For example, in situations where cpu_user and cpu_sys metrics are low values such as 5%, the cpu_idle value can be 190%. This issue is resolved in release 2.0.0.1.

Upgrading to Greenplum Performance Monitor 2.0.0.1

To upgrade an existing version 2.0.0.0 instance of Greenplum Performance Monitor, you must install the new program files using the 2.0.0.1 installer. This patch does not require any further updating of monitoring components.

To upgrade from version 2.0.0.0

1. Download or copy the installer file to the Greenplum Database master host. Installer files are available from Greenplum for RedHat (32-bit and 64-bit) and Solaris (64-bit) platforms.

2. Unzip the installer file where *PLATFORM* is either `RHEL4-i386` (RedHat 32-bit), `RHEL4-x86_64` (RedHat 64-bit) or `SOL-x86_64` (Solaris 64-bit). For example:

   ```
   # unzip greenplum-perfmon-2.0.0.1-PLATFORM.zip
   ```

3. Stop Greenplum Performance Monitor. For example:

   ```
   $ gpperfmon --stop
   ```

4. Launch the installer using `bash`. For example:

   ```
   # /bin/bash greenplum-perfmon-2.0.0.1-PLATFORM.bin
   ```

5. The installer will prompt you to accept the Greenplum Performance Monitor license agreement. Type `yes` to accept the license agreement.

6. The installer will prompt you to provide an installation path. Enter the location of your existing performance monitor installation. This is the location referred to by the variable `$GPPERFMONHOME`.

7. The installation directory contains the file `gpperfmon_path.sh` with path and environment settings for the monitor. You must source this file in your `gpadmin` user's startup shell profile (such as `.bashrc`).

   For example, you could add a line similar to the following to your chosen profile files:

   ```
   source /usr/local/greenplum-perfmon-2.0.0.0/gpperfmon_path.sh
   ```

8. Restart Greenplum Performance Monitor. For example:

   ```
   $ gpperfmon --restart
   ```

Greenplum Performance Monitor 2.0.0.0

Greenplum Performance Monitor 2.0.0.0 is compatible with Greenplum Database 3.2.1.0 or later. If you are running an earlier release of Greenplum Database, you must upgrade to version 3.2.1.0 in order to use Greenplum Performance Monitor. For more information on upgrading, see the *Greenplum Database 3.2.1.0 Release Notes*.

Known Issues in Greenplum Performance Monitor 2.0.0.0

This section lists the customer reported issues that remain unresolved in Greenplum Performance Monitor 2.0.0.0. Workarounds are provided where applicable.

Table A.2 Known Issues in Greenplum Performance Monitor

Issue Number	Description
4540	**Console intermittently fails to refresh display** In rare cases, Greenplum Performance Monitor console running on Firefox may experience a caching issue that stops the updating of metrics information in the display. Workaround: close the browser and log on to Greenplum Performance Monitor console in a new browser window.
4516	**Iterator data may display inaccurate shared memory values** In the metrics for a given iterator, the value for *memshare* may be recorded as `-1`. This issue affects the *memshare* column of *iterators_now*, *iterators_tail*, and *iterators_history* tables. It also affects the **Shared mem** field displayed for iterators in the Greenplum Performance Monitor console.
4439	**Missing or inaccurate performance data for some scalar subqueries** Performance data for some scalar subqueries may have missing iterator data. Also, the query start time may be shown as the time that subquery is completed.
3936	**Monitor agent attempts to connect on reserved port** If the parameter `gpperfmon_port` is configured to use a port already in use, the master monitor agent cannot connect to segment agents to harvest data. However, the master agent starts normally without an error message, and continues to attempt to connect to segment agents.
3913	**Value for cpu_elapsed not correctly logged for short-running queries** When a query runs for a period shorter than `quantum` (default value is 15 seconds), the value for cpu_elapsed may be logged as zero. This may occur even though the query runtime is greater than the values for `min_query_time` and `min_detailed_query`, when those values are lower than the value for `quantum`.

Installing Greenplum Performance Monitor 2.0.0.0

These are the high-level steps to install Greenplum Performance Monitor. For detailed instructions, please see the installation chapter of the *Greenplum Performance Monitor Guide*.

1. In your Greenplum Database 3.2.1 system, create the `gpmon` superuser role.
2. Enable database access for the `gpmon` superuser by configuring client authentication privileges in `pg_hba.conf`. For example, to allow trusted access both locally and from an interface with the IP address 10.0.2.1, add the following entries to this file

   ```
   local      gpperfmon      gpmon      trust
   host       gpperfmon      gpmon     10.0.2.1/32     trust
   ```

3. Unless you explicitly allow trust login, password authentication is required for all database users that will be accessing the Greenplum Performance Monitor console. Make sure that all administrators that will log in to the Greenplum Performance Monitor have encrypted database passwords. To enforce this, add an entry like the following to `pg_hba.conf`:

   ```
   host    all     gpadmin     0.0.0.0/0     md5
   ```

After you have created entries for both the `gpmon` user and the `gpadmin` user, `pg_hba.conf` will include a section like the following:

```
# Entries for the gpmon superuser
local     gpperfmon     gpmon     trust
host      gpperfmon     gpmon    10.0.2.1/32    trust
# Entry for the gpadmin user
host     all           gpadmin    0.0.0.0/0     md5
```

Note that the the `gpmon` user cannot log in to the Greenplum Performance Monitor console with these settings. Log in to the console using `gpadmin`.

4. Add the monitor `postgresql.conf` parameters on the master host. The parameter `gp_enable_gpperfmon=on` is required for query metric collection. After editing `postgresql.conf`, reload the configuration using `gpstop -u`.
5. Run the Greenplum Performance Monitor installer on the Greenplum Database master host. After installation, source the `gpperfmon_path.sh file` in your `gpadmin` user profile file.
6. Copy your Greenplum Performance Monitor installation to your segment hosts.
7. Set up the monitor database, master agent and web service on the master host using the `gpperfmon` utility.

Greenplum Performance Monitor 2.0.0.0 Documentation

The following Greenplum Performance Monitor documentation is available in the `$GPPERFMONHOME/docs` directory of your Greenplum Performance Monitor installation, or you can go to http://gpn.greenplum.com to download the latest documentation:

GPMonGuide.pdf - *Greenplum Performance Monitor Guide*